

Assignment 3

1. Find the critical radius of systems composed of U-235 and Pu-239, each surrounded by a reflector made of:

- (a) water,
- (b) iron,
- (c) sodium.

For each reflector material, consider reflector optical thicknesses of

$$1 \text{ mfp}, \quad 2 \text{ mfp}, \quad 3 \text{ mfp}, \quad 10 \text{ mfp}.$$

Use the cross-section data given in Question 5 of Assignment 1.

Perform the calculation using the following approximations:

- (a) the continuous diffusion approximation,
- (b) the continuous asymptotic diffusion approximation,
- (c) the discontinuous asymptotic diffusion approximation.

Compare the results with those obtained for the corresponding bare systems in Question 5 of Assignment 1.

Hint for Questions 2–5: Determine the critical system size by varying the system size until the multiplication factor satisfies $k = 1$. A bisection or Newton–Raphson algorithm may be used to locate the critical system size.

2. Find the critical half-thickness in slab geometry using the P_N approximation.

Write a numerical code and use an iterative k -eigenvalue algorithm to determine the critical half-thickness for

$$c = 1.2, \quad c = 1.5, \quad c = 1.8.$$

Present the results for

$$N = 1, \quad N = 3, \quad N = 5, \quad N = 9.$$

3. Find the critical half-thickness in slab geometry using the S_N approximation.

Write a numerical code and use an iterative k -eigenvalue algorithm to determine the critical half-thickness for

$$c = 1.2, \quad c = 1.5, \quad c = 1.8.$$

Present the results for

$$N = 2, \quad N = 4, \quad N = 6, \quad N = 10.$$

Use a Gauss–Legendre quadrature set, the diamond-difference spatial discretization scheme, and a negative-flux fixup.

Compare the P_N and S_N results, and explain the possible source of the difference between P_N and S_{N+1} for low values of N .

4. Find the critical radius in spherical geometry using the S_N approximation.

Write a numerical code and use an iterative k -eigenvalue algorithm to determine the critical radius for

$$c = 1.2, \quad c = 1.5, \quad c = 1.8.$$

Present the results for

$$N = 2, \quad N = 4, \quad N = 6, \quad N = 10.$$

Use a Gauss–Legendre quadrature set, the diamond-difference spatial discretization scheme, and a negative-flux fixup.

5. Find the critical mass of U-235 and Pu-239 using the S_N method.

Use the cross-section data given in the table in Question 5 of Assignment 1.

Write a numerical code and use an iterative k -eigenvalue algorithm to determine the critical system size of each material.

Determine the corresponding critical mass of each system.